$$= \sigma^2 + \mu^2 - \left(\mathrm{Var}(\bar{x}) + \mathbb{E}[\bar{x}]^2\right)$$

$$= \sigma^2 + \mu^2 - \left(\mathrm{Var}\left(\tfrac{1}{k}\textstyle\sum x_i\right) + \mathbb{E}\left[\tfrac{1}{k}\textstyle\sum x_i\right]^2\right)$$

$$= \sigma^2 + \mu^2 - \left(\tfrac{1}{k^2}\textstyle\sum \mathrm{Var}(x_i) + \tfrac{1}{k^2}\mathbb{E}[\textstyle\sum x_i]^2\right)$$

$$= \sigma^2 + \mu^2 - \tfrac{1}{k^2}\left(k\mathrm{Var}(x) + k^2\mathbb{E}[x]^2\right)$$

$$= \sigma^2 + \mu^2 - \frac{\sigma^2}{k} - \mu^2$$

$$= \sigma^2 - \frac{\sigma^2}{k} = \frac{k\sigma^2 - \sigma^2}{k} = \sigma^2\left(\frac{k-1}{k}\right.$$